

Getting the Riemann Curvature Tensor

$$(\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) V^\mu = \nabla_\alpha (V^\mu_{;\beta} + \Gamma^\mu_{\sigma\beta} V^\sigma) - (\alpha \leftrightarrow \beta)$$

(1) tensor $\rightarrow$ what is this V^μ ?
 Cov deriv takes tensor into tensors. $V^\mu_{;\beta}$

$$= (\quad)_{;\alpha} + \Gamma^\mu_{\tau\alpha} (\quad)^{\tau\mu} - \Gamma^\tau_{\beta\alpha} (\quad)_{\beta\tau} - [\alpha \leftrightarrow \beta]$$

$$= \cancel{V^\mu_{;\beta\alpha}} + \Gamma^\mu_{\sigma\beta,\alpha} V^\sigma + \Gamma^\mu_{\sigma\beta} V^\sigma_{;\alpha}$$

$$+ \Gamma^\mu_{\tau\alpha} (\cancel{V^\tau_{;\beta}} + \Gamma^\tau_{\sigma\beta} V^\sigma)$$

$$- \Gamma^\tau_{\beta\alpha} (\cancel{V^\mu_{;\tau}} + \Gamma^\mu_{\sigma\tau} V^\sigma)$$

$$- \cancel{V^\mu_{;\alpha\beta}} - \Gamma^\mu_{\sigma\alpha,\beta} V^\sigma - \Gamma^\mu_{\sigma\alpha} V^\sigma_{;\beta}$$

$$- \Gamma^\mu_{\tau\beta} (\cancel{V^\tau_{;\alpha}} + \Gamma^\tau_{\sigma\alpha} V^\sigma)$$

$$+ \Gamma^\tau_{\alpha\beta} (\cancel{V^\mu_{;\tau}} + \Gamma^\mu_{\sigma\tau} V^\sigma)$$

characterizes the 2nd derivative of the metric
 The Riemann Curvature Tensor

$$= (\Gamma^\mu_{\sigma\beta,\alpha} - \Gamma^\mu_{\sigma\alpha,\beta} + \Gamma^\mu_{\tau\alpha} \Gamma^\tau_{\sigma\beta} - \Gamma^\mu_{\tau\beta} \Gamma^\tau_{\sigma\alpha}) V^\sigma = R^\mu_{\sigma\alpha\beta} V^\sigma$$

(1) R is a tensor

$$(2) R_{\alpha\beta\gamma\delta} = g_{\alpha\mu} R^\mu_{\beta\gamma\delta}$$

$$(a) R_{\alpha\beta\gamma\delta} = -R_{\beta\alpha\gamma\delta}$$

$$(b) R_{\alpha\beta\gamma\delta} = -R_{\alpha\beta\delta\gamma}$$

$$(c) R_{\alpha\beta\gamma\delta} = R_{\gamma\delta\alpha\beta}$$

$$(d) R_{\alpha\beta\gamma\delta} + R_{\alpha\delta\gamma\beta} + R_{\alpha\gamma\delta\beta} = 0$$

(e) Bianchi Identity

R has $\frac{d^2(d^2-1)}{12}$ indep. components.

Try to prove these symmetries of $R \rightarrow g^{\mu\nu}$ vanishes

Let x^μ such that $R^\mu_{\sigma\alpha\beta} = \frac{1}{2} [g^{\mu\tau} (g_{\tau\sigma,\beta} + g_{\tau\beta,\sigma} - g_{\sigma\beta,\tau})]_{,\alpha}$
 $g_{\alpha\beta,\gamma} = 0$

$$- \frac{1}{2} [g^{\mu\tau} (g_{\tau\sigma,\alpha} + g_{\tau\alpha,\sigma} - g_{\sigma\alpha,\tau})]_{,\beta}$$

$$R_{\epsilon\sigma\alpha\beta} = \frac{1}{2} [g_{\epsilon\sigma,\beta\alpha} + g_{\epsilon\beta,\sigma\alpha} - g_{\sigma\beta,\epsilon\alpha}]$$

$$- \frac{1}{2} [g_{\epsilon\sigma,\alpha\beta} + g_{\epsilon\alpha,\sigma\beta} - g_{\sigma\alpha,\epsilon\beta}]$$

$g_{\mu\epsilon}$

$$R_{\alpha\beta\gamma\delta} = \frac{1}{2} (g_{\alpha\delta,\beta\gamma} - g_{\beta\delta,\alpha\gamma} - g_{\alpha\gamma,\beta\delta} + g_{\beta\gamma,\alpha\delta})$$

Branchi Identity:

$$R_{\alpha\beta\gamma\delta} \delta_{\mu}^{\gamma} + R_{\alpha\beta\delta\mu} \delta_{\gamma}^{\delta} + R_{\alpha\beta\mu\gamma} \delta_{\delta}^{\gamma} = 0$$

Proof: In the special point where $g_{\alpha\beta,\mu} = 0 \Rightarrow ; \rightarrow ,$

$$= \frac{1}{2} (g_{\alpha\delta,\beta\gamma\mu} + g_{\beta\gamma,\alpha\delta\mu} - g_{\beta\delta,\alpha\gamma\mu} - g_{\alpha\gamma,\beta\delta\mu}$$

+

$$= 0$$

$$\underline{R_{\alpha\beta} \equiv g^{\mu\nu} R_{\mu\alpha\nu\beta} \text{, "Ricci Tensor"}}$$

$$\underline{\text{Curvature } \overset{\text{Scalar}}{\text{Tensor}} \quad R \equiv g^{\alpha\beta} R_{\alpha\beta} .}$$

$$(R_{\alpha\beta\gamma\delta} + R_{\alpha\beta\delta\gamma} + R_{\alpha\beta\gamma\delta}) g^{\alpha\delta} g^{\beta\gamma} = 0$$

$$g^{\beta\mu} R_{\beta\delta\gamma\mu} + R_{\alpha\delta\gamma\alpha} g^{\alpha\gamma} - R_{\beta\mu\gamma\delta} g^{\beta\mu} = 0$$

$$2R^{\mu}_{\delta\gamma\mu} - R_{,\delta} = 0$$

$$\nabla^{\mu} (R_{\delta\mu} - \frac{1}{2} g_{\delta\mu} R) = 0$$

$$\underbrace{\hspace{10em}}_{G_{\delta\mu}} \Rightarrow \nabla^{\alpha} G_{\alpha\beta} = 0$$

$$R_{\delta\mu;\mu} - \frac{1}{2} g_{\delta\mu;\mu} R - \frac{1}{2} g_{\delta\mu} R_{;\mu}$$